

SECTION 4: RAG SYSTEM OBSERVATION REPORT

1-Page Synthesis & Learning Assessment Report

Student Name: SHRI RAM PRINCE MISHRA | Application No.: IN26012152 | GitHub:
https://github.com/srpm2005/HelpDeskManagement_IN26012152

Q1: Was the AI able to answer questions that were available in your project documentation?

Yes, absolutely. When queried with in-domain questions (such as API endpoints, application architecture, EF Core database models, unit test setups, and validation rules), the RAG Knowledge Assistant successfully retrieved the relevant document chunks with **100% precision (10/10 test cases passed)**. It produced clear, structured answers backed by exact source file paths and line-number citations (e.g. README.md#L81-L93 and TicketService.cs#L1-L93).

Q2: How did the AI respond when the information was not available in the document?

When presented with out-of-domain general knowledge questions absent from the codebase documentation (e.g., *Who is the CEO of Microsoft?*, *What is Artificial Intelligence?*, *What is the capital of Australia?*), the RAG system strictly enforced grounding controls. Instead of hallucinating or making up speculative answers, the AI responded consistently with:
"The uploaded project documentation does not contain this information."

Q3: What did you learn about Retrieval-Augmented Generation (RAG)?

Key learnings from building and evaluating the RAG Knowledge Assistant:

- **Solves Hallucination & Staleness:** Pre-trained LLMs lack access to private enterprise codebases. RAG dynamically injects relevant private context at query time without expensive fine-tuning.
- **Provenance & Explainability:** Unlike standard LLM answers, RAG provides explicit file and line-number citations, allowing developers to verify source truth instantly.
- **Chunking & Retrieval Strategy:** Smart header-aware markdown chunking and code-aware block parsing, combined with hybrid vector similarity search (TF-IDF + keyword boosting), achieve microsecond retrieval latencies (< 2 ms).

Q4: Mention at least three real-world software applications where RAG can be useful.

1. **Developer Onboarding & Internal Codebase Q&A:** Software engineering teams can deploy RAG assistants over repository documentation, pull requests, and architectural decision records (ADRs) to onboard new developers seamlessly.
2. **Enterprise Legal & Compliance Contract Analysis:** Legal departments can query thousands of private contracts and regulatory policies to extract clauses and risk terms with exact document citations.
3. **Customer Support & Technical HelpDesk Systems:** Customer service agents can utilize RAG over proprietary product manuals and troubleshooting guides to generate accurate resolution steps instantly.

Visual Proof: Grounded RAG Response & Context Inspector

